

Cheng-Yuan Li

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science, GPA: 4.0/4.0

Atlanta, GA

Aug. 2024 – May 2028

TECHNICAL SKILLS

Languages: Python, Go (Golang), SQL (PostgreSQL, MySQL, Redis), C++, C, Java

Data Systems & Architecture: Distributed Systems, Key-Value Storage, LSM-Trees, Query Optimization, Concurrency & Caching, Log-Structured Engines, Message Passing (SQS, Kafka concepts), Async I/O, REST APIs

Cloud, Infrastructure & Tooling: AWS (S3, SQS), Linux, Docker, Terraform & Kubernetes concepts, Agile/Scrum, CI/CD, Version Control (Git), TDD

WORK EXPERIENCE

Backend Software Engineer Intern: Data Systems

May 2026 – Aug. 2026

Gordon Food Service, Inc.

Atlanta, GA

- Engineered an automated serverless data-cost management service using REST APIs and BigQuery slot allocation, securing **\$50,000+ in annual infrastructure savings**.
- Refactored a 7 TB weekly analytical data engine with an incremental merge pipeline using deterministic key hashing, **reducing data scan overhead by 97.8%** (7 TB to 150 GB).
- Architected an asynchronous log-parsing pipeline processing 3 TB of execution traces across 4,000+ job configurations on Linux environments, eliminating execution latency and pipeline failures.

Teaching Assistant: Database Systems & Object-Oriented Software

Jan. 2025 – Present

Georgia Institute of Technology

Atlanta, GA

- **CS 4400 (Database Systems):** Mentor 100+ students on PostgreSQL schema design, B+ tree indexing, ACID transaction isolation, and SQL query optimization for high-concurrency systems.
- **CS 1331 (Introduction to OOP):** Lead labs for 220+ undergraduates, enforcing clean software architecture, object-oriented design patterns, REST API principles, and automated TDD (JUnit).

Research Intern – Distributed Data Performance

May 2025 – Aug. 2025

*A*STAR Institute of High Performance Computing*

Singapore

- Built an automated profiling framework to benchmark execution throughput, memory footprints, and compute latency across 40+ large-scale data system architectures.
- Engineered log-processing pipelines to analyze multi-gigabyte device execution traces, pinpointing memory bandwidth constraints and optimizing cache utilization.

PROJECTS

SmallDB: High-Performance LSM-Tree Database | C++

Aug. 2025 – Present

- Architected a crash-consistent key-value storage engine from scratch in C++, achieving **52,874 write ops/sec** at 18.91 μ s latency via an LSM-tree memtable, append-only WAL, and background thread compaction.
- Increased read query throughput by **89x (174 to 15,515 ops/sec)** by engineering Bloom filter indexes, sparse block indexing, and a fixed-size concurrent block cache to minimize disk I/O.
- Eliminated write stalls during high-concurrency bursts using a streaming min-heap SSTable compaction worker while maintaining **sub-30ms WAL crash recovery**.

Distributed Data Lakehouse & Streaming Pipeline | Python, AWS, Databricks

Aug. 2026 – Present

- Engineered an async data ingestion client (`aihttp`, `asyncio`) featuring client-side RSA-PSS request signing and an in-memory token-bucket rate limiter with jittered backoff.
- Designed an incremental Hive-partitioned JSONL landing layer with status-diff tracking, preserving out-of-order event delivery while maintaining transaction state consistency.
- Constructed a cloud data mirror syncing Bronze partitions to **AWS S3**, implementing PySpark ETL pipelines for deduplication, schema enforcement, and compaction on **Databricks**.